Functional Skills
Mathematics Level 1

Practice

OFFICIAL MFS MODEL SOLUTION

60 / 60 Marks (100% Pass)

- Assessment: Practice Assessment 2 (FSM109P)
- Complete step-by-step worked models
- Official Mark Scheme Allocations [M1 / A1]
- Exam Board: Open Awards (Ofqual Regulated)

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LEVEL 1 FUNCTIONAL SKILLS QUALIFICATION IN MATHEMATICS

SECTION A - QUESTION AND ANSWER PAPER
NON-CALCULATOR – 30 MINUTES
PRACTICE ASSESSMENT 1 (FSM109P)

Do not open this paper until you are told to do so by the invigilator.

Overall assessment marks available: **60**
Overall assessment time limit: **2 HOURS**

There are **TWO** Sections to this assessment:

- **Section A** includes Task 1. You **must not** use a calculator for this section.

  Total marks available: 15. Time limit: 30 minutes
- **Section B** includes Task 2, 3 and 4. You can use a non-scientific calculator for this section

  Total marks available: 45. Time limit: 1 hour and 30 minutes

For Section A you need:

- This question and answer paper
- A pen with black or blue ink
- A pencil
- A ruler and a protractor

INTERNET ACCESS IS NOT PERMITTED AND YOU MUST NOT USE A CALCULATOR

The invigilator will stop the assessment after 30 minutes. You must hand in this question and answer paper at this point.

The invigilator will then hand out **Section B** and a non-scientific calculator. You will then have a further 1 hour and 30 minutes to complete **Section B**.

Instructions

1. Please sign and date below to confirm that your details are correct and that you have understood the instructions.
2. Read each task and question carefully.
3. Remember to show all your workings out clearly.
4. The number of marks available for each question is shown in brackets. Use these marks to guide you on how long to spend on each question.
5. Answer **all** questions using the space provided on this question and answer paper.
6. If you have time, check your work for **Section A** at the end. Once you have handed in this question and answer paper, you will not be able to check this again.
7. If you use extra paper, write your name, learner number and the question number you are answering on it and securely attach it to this question and answer paper.

Learner name:	
Learner number:	
Centre number:	
Signature:	
Today's date:	

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For Markers Use Only

Section A

Question 1

Calculate $-10 + 3$ (1 mark)

Evaluate $5^2 + 1$:

Show your calculations and/or workings out here: [1 mark]

$5^2 = 5 \times 5 = 25 \rightarrow 25 + 1 = 26$

26 ✓

Write your answer in this box:

Identify fraction shaded on grid:

3 parts shaded out of 8 total parts = 3/8 [1 mark]

Question 2

Calculate 1000×0.15 (1 mark)

Show your calculations and/or workings out here:

3/8 ✓

Write your answer in this box:

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Question 3

Calculate 45.53 – 24.37 (1 mark)

Subtract decimals aligning decimal point:

Show your calculations and/or workings out here:

45.8 - 29.4 = 16.4 [1 mark]

16.4 ✓

Write your answer in this box:

Question 4

Calculate 105 – 25 × 2 (1 mark)

Show your calculations and/or workings out here:

Multiply by 100 (shift decimal point 2 places right):

5.26 × 100 = 526 [1 mark]

526 ✓

Write your answer in this box:

For Markers Use Only

Question 5

Which of the following percentages is equivalent to $\frac{2}{5}$? (1 mark)

Convert 3/5 to percentage:

10% 20% 25% 40% 80%

3/5 = 6/10 = 60/100 = 60%

[1 mark]

Show your calculations and/or workings out here:

60% ✓

Write your answers in this box:

Question 6

Order temperatures from coldest to warmest:

-5°C < -2°C < 0°C < 3°C < 7°C

[1 mark]

Put the following fractions in order of size: (1 mark)

$\frac{3}{5}$ $1\frac{1}{4}$ $\frac{1}{2}$ $1\frac{1}{5}$ $\frac{2}{3}$

Show your calculations and/or workings out

-5, -2, 0, 3, 7 ✓

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Question 7

Sasha is an art teacher.

She buys art materials in packs and then sells them on individually to her students at cost price.

The prices she pays for the packs of items are shown in the following table:

Item	Quantity	Cost price
Paint brushes	Pack of 100	£39.00
Canvas board	Pack of 10	£18.50
Tubes of paint	Pack of 10	£14.29

Step 1: Calculate cost of each item type:

• 3 Notebooks @ £4.50 = £13.50

• 2 Packs of pens @ £2.85 = £5.70

• 1 Folder @ £3.75 = £3.75

A student buys ten paint brushes, three tubes of paint and one canvas board. [M1 A1]

How much in total should Sasha charge the student? (4 marks)

Step 2: Add all items together:

Show your calculations and/or workings out here:

£13.50 + £5.70 + £3.75 = £22.95 [M1 A1]

£22.95 ✓

Write your answer in this box:

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Question 8

Sasha wants to make orange paint by mixing red and yellow paint in the ratio 3:2

She uses 180ml red paint.
She has 100ml of yellow paint.
Has Sasha got enough yellow paint? (2 marks)

Show your calculations and/or workings out here:

Step 1: Read scale interval and calculate remaining volume:

Capacity = 250 ml, Level indicates 75 ml used

250 ml - 75 ml = 175 ml remaining [M1 A1]

Write your answer in this box:

175 ml ✓

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Question 9

Sasha needs to put tape around the edge of five paintings.

The paintings are rectangular and are 48.9cm long and 32.6cm wide.

Estimate how much tape she needs altogether for the 5 paintings.

(3 marks)

Show your calculations and/or workings out here:

Step 1: Round measurements to nearest 10 for estimation:

Length 48cm --> 50cm, Width 31cm --> 30cm [M1]

Step 2: Perimeter per frame = (50 + 30) × 2 = 160 cm [M1]

Step 3: Tape needed for 5 paintings = 5 × 160 cm = 800 cm [A1]

800 cm ✓

Write your answer in this box:

[End of Section A]

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LEVEL 1 FUNCTIONAL SKILLS QUALIFICATION IN MATHEMATICS

SECTION B - QUESTION AND ANSWER PAPER
CALCULATOR – 1 HOUR 30 MINUTES
PRACTICE ASSESSMENT 1 (FSM109P)

Do not open this paper until you are told to do so by the invigilator.

Overall assessment marks available: **60**
Overall assessment time limit: **2 HOURS**

There are **TWO** Sections to this assessment:

- **Section A** – please ensure you have handed in Section A before beginning Section B
- **Section B** includes Task 2, 3 and 4. You can use a non-scientific calculator for this section.

Total marks available: 45. Time limit: 1 hour and 30 minutes.

For Section B you need:

- This question and answer paper
- A pen with black or blue ink
- A pencil
- A ruler and a protractor
- A non-scientific calculator

INTERNET ACCESS IS NOT PERMITTED

You now have a further 1 hour and 30 minutes to complete **Section B**.

Instructions

1. Please sign and date below to confirm that your details are correct and that you have understood the instructions.
2. Read each task and question carefully.
3. Remember to show all your workings out clearly.
4. The number of marks available for each question is shown in brackets. Use these marks to guide you on how long to spend on each question.
5. Answer **all** questions using the space provided on this question and answer paper.
6. If you have time, check your work for **Section B** at the end.
7. If you use extra paper, write your name, learner number and the question number you are answering on it, and securely attach it to this question and answer paper.
8. At the end of this section (**Section B**), hand in this question and answer paper and all notes to the invigilator.

Learner name:	
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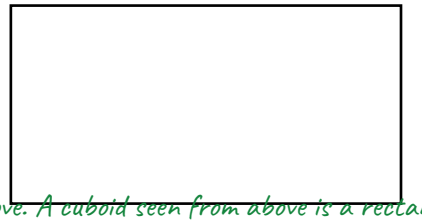
Section B

Question 10



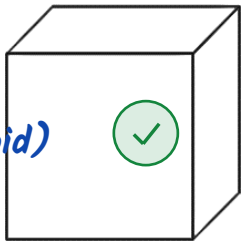
Plan view looks directly from above. A cuboid seen from above is a rectangle.

Plan view

Shape D is a cuboid. [1 mark]

The diagram above shows the plan view of which shape below? (1 mark)

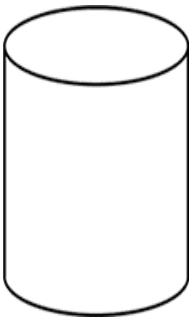
A



D (Cuboid)

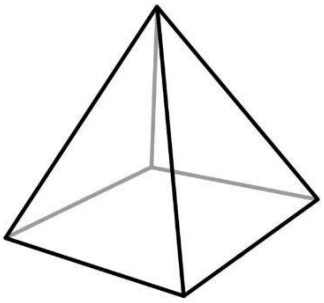
Cube

B



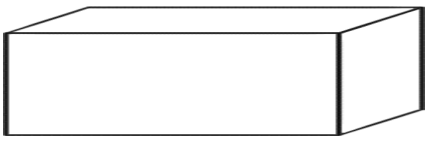
Cylinder

C



Square-based pyramid

D



Cuboid

Write your answer in this box:

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Question 11

The table below shows the ages of 18 visitors to a museum.

2	4	28	35	19	21
7	31	35	48	10	41
38	8	9	27	18	25

Complete the table below. (1 mark)

Only need one age in each 10-year bracket:

0–9: 5, 10–19: 3, 20–29: 4, 30–39: 4, 40–49: 2 *[1 mark]*

Age (years)	Frequency
0–9	
10–19	
20–29	
30–39	
40–49	

Show your calculations and/or workings out here:

Frequencies: 5, 3, 4, 4, 2 (Total = 18) ✓

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Question 12

Which of the following percentages is the highest? (1 mark)

37% 42% 85% 87% 56% 57% 18%

Compare percentage values:

87% is the highest percentage. [1 mark]

Write your answer in this box:

87% ✓

Question 13

Eilish owns a children's soft-play area.

She has a ball pit which is 2.4m by 2.4m and has a depth of 0.45m.

The ball pit needs 2250 balls for each cubic metre (m^3).

How many balls will it take to fill the ball pit? (4 marks)

Volume of ball pit: Length × Width × Depth:

Volume = 2.4 × 2.4 × 0.45 = 2.592 m^3 [M1 A1]

Number of balls = 2.592 × 2250 = 5832 balls [M1 A1]

5832 balls ✓

Show your calculations and/or workings out here:

Write your answer in this box:

For Markers Use Only

Question 14

Eilish has 30 mats to cover the floor of the soft-play area.

The soft-play floor mats come in three colours: black, light grey and dark grey.

$\frac{1}{5}$ of the soft-play floor mats are black.

There are equal numbers of patterned mats.

Step 1: 30 mats in total. 1/5 are black:

On the diagram of the floor seen below, arrange the mats in a pattern with at least one line of symmetry.

30 ÷ 5 = 6 black mats [M1]

(3 marks)

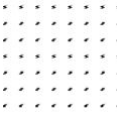
Step 2: Remaining 24 mats split equally into light and dark grey:

Floor (30 mat spaces)

12 light grey and 12 dark grey mats [A1]

Arranged symmetrically around vertical centre line				

Diagonal

Dots 

Grey

[1 mark]

Show your calculations and/or workings out here:

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Question 15

The soft-play area can be hired for children's parties. The prices are as follows:

£90 to hire the soft play area for up to 40 children
£17 for party food for 5 children
£12 for 10 party bags
15% discount for parties Monday to Thursday

Eilish takes a booking for a party of 30 children for a Thursday including food and a party bag for each child.

How much should she charge for the soft-play party for 30 children including food and party bags for each child?

(5 marks)

Show your calculations and/or workings out here:

Step 1: Calculate total cost before discount:

- Food: 30 ÷ 5 × £17 = £102
- Party bags: 30 ÷ 10 × £12 = £36
- Room hire: £90

Total = £102 + £36 + £90 = £228 [M1 A1]

Step 2: Apply 15% discount (pay 85%):

Discount = £228 × 0.15 = £34.20

Final price = £228 − £34.20 = £193.80 [M1 A1]

£193.80 ✓

Write your answer in this box:

For Markers Use Only

Question 16

Write 832304 in words. (1 mark)

Write 832,304 in words:

Write your answer in this box:

Eight hundred and thirty-two thousand, three hundred and four [1 mark]

Eight hundred and thirty-two thousand, three hundred and four

Question 17

There are three red sweets, four yellow sweets and three blue sweets in a bag.
What is the probability that a sweet taken out of the bag at random is red?
Give your answer as a fraction.

(1 mark)

Show your calculations and/or workings out here:

Probability = Successful outcomes ÷ Total outcomes:

3 winning tickets out of 10 total = 3/10 [1 mark]

3/10

Write your answer in this box:

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Question 18

Calculate the range of this set of numbers. (1 mark)

23.1 25.8 25.3 23.6 19.1 21.2

Show your calculations and/or workings out here:

Range = Highest value - Lowest value:

25.8 - 19.1 = 6.7°C [1 mark]

Write your answer in this box:

6.7°C

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Question 19

Amy is 36 years old and wants to improve her fitness level.

She uses this formula to find the healthy upper and lower heart rate limits, in beats per minute (bpm), for her age during moderate exercise.

Lower heart rate = (220 – age) × 0.6
Upper heart rate = (220 – age) × 0.7

Step 1: Calculate target heart rate range for age 36:

Her heart rate while exercising is 118 bpm.

Lower limit: (220 – 36) × 0.60 = 184 × 0.60 = 110.4 bpm

Upper limit: (220 – 36) × 0.70 = 184 × 0.70 = 128.8 bpm [M1 A1]

During exercise, is her heart rate **within** the healthy upper and lower limits for her age?

Step 2: Heart rate was 120 bpm, which is between 110.4 and 128.8 bpm. [A1] (3 marks)

Show your calculations and/or workings out here:

Yes ✓

Target range is 110.4 to 128.8 bpm; 120 bpm is within range.

Write your answer in this box:

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Question 20

Amy records how much exercise she completed during the past week:

¾ hour swimming on both Monday and Thursday

1¼ hours in the gym on Saturday

20 minutes brisk walking on five days

How much time did she spend exercising altogether over the week?
Give your answer in hours and minutes. (2 marks)

Step 1: Sum time intervals in minutes:

45 + 45 + 75 + (20 × 5) = 165 + 100 = 265 minutes [M1]

Show your calculations and/or workings out here:

Step 2: Convert to hours and minutes (265 ÷ 60 = 4 r 25):

265 minutes = 4 hours and 25 minutes [A1]

Write your answer in this box:

4 hours 25 minutes

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Question 21

Amy walks 6680 steps on Monday.

Each step length is an average of 0.7m.

Using her average step length, how many kilometres does she walk on Monday?

Give your answer to one decimal place. (4 marks)

Step 1: Calculate total distance in metres:

6680 steps × 0.7m per step = 4676 metres [M1 A1]

Show your calculations and/or workings out here:

Step 2: Convert metres to kilometres (÷ 1000):

4676 ÷ 1000 = 4.676 km [M1]

Step 3: Round to 1 decimal place: 4.676 --> 4.7 km [A1]

4.7 km ✓

Write your answer in this box:

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Question 22

Amy records her blood pressure over the week.

These are her blood pressure values:

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Blood pressure	113	121	99	112	118	115	129

Step 1: Add the 5 weekday values together:

113 + 121 + 112 + 118 + 115 = 579 [M1]

She has been told that to calculate her mean blood pressure she should remove the highest and lowest values first.

What is Amy's mean blood pressure? (3 marks)

Step 2: Divide sum by number of days (5):

Mean = 579 ÷ 5 = 115.8 [M1 A1]

Show your calculations and/or workings out here:

Write your answer in this box:

115.8 ✓

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Question 23

The table below shows the number of parcels delivered by a delivery driver over four weeks.

Weeks	1	2	3	4
Number of parcels	430	560	720	850

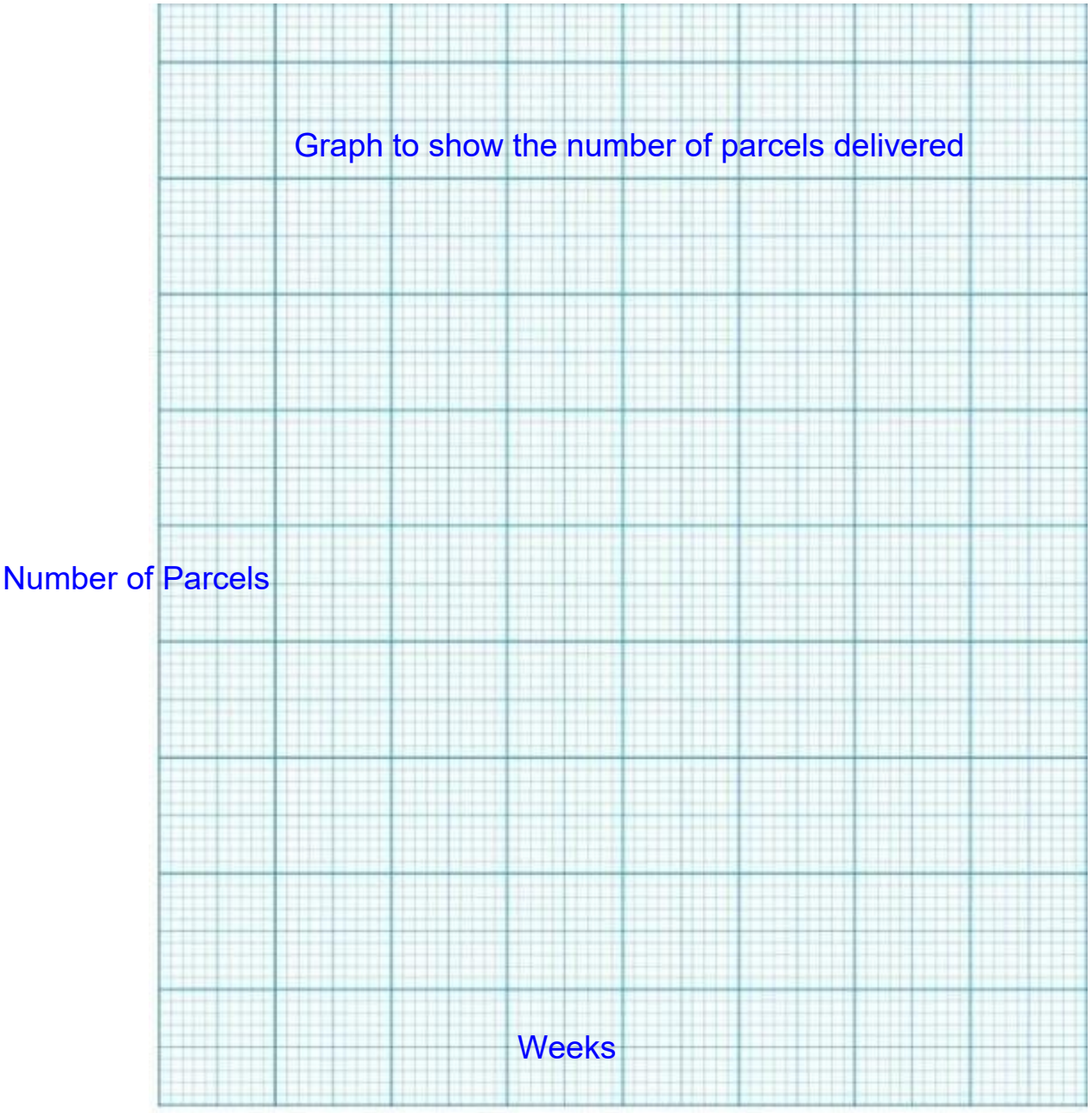
Draw a line graph to show the number of deliveries over the four weeks.

Line Graph Checklist:

- X-axis: Weeks 1, 2, 3, 4
- Y-axis: Number of parcels delivered (scale 0 to 140 in 20s)
- Accurate plotted points connected by straight line segments

(3 marks)

[3 marks]


Graph to show the number of parcels delivered
Number of Parcels
Weeks

For Markers Use Only

Question 24

Liam is a student who has worked over the summer.

He has earned £2304.26 altogether.

He earns £8.72 per hour.

He is entitled to 0.12 hours of holiday for every hour he has worked.

How many hours holiday is he entitled to? (4 marks)

Show your calculations and/or workings out here:

Step 1: Calculate total hours worked (Total pay ÷ hourly rate):

2304.26 ÷ 8.72 = 264.25 hours [M1 A1]

Step 2: Calculate 12% holiday entitlement on worked hours:

Holiday hours = 264.25 × 0.12 = 31.71 hours [M1 A1]

Write your answer in the box:

31.71 hours ✓

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Question 25

Liam earned £2304.26 altogether.

He spent £399 on a new laptop and puts half of his remaining earnings into a savings account.

The savings account pays 5% interest per year.

How much money will Liam have in his savings account after one year?

(3 marks)

Show your calculations and/or workings out here.

Step 1: Calculate remaining income saved equally between 2 accounts:

(£2304.26 − £399) ÷ 2 = £1905.26 ÷ 2 = £952.63 [M1]

Step 2: Add 5% annual interest (× 1.05):

Interest = £952.63 × 0.05 = £47.63

Total in account = £952.63 + £47.63 = £1000.26 [M1 A1]

£1000.26 ✓

Write your answer in this box:

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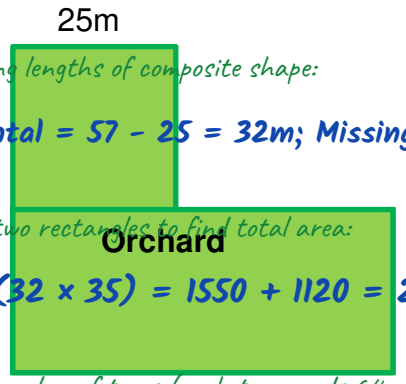
Question 26

Anya is planting some fruit trees in her orchard.

The dimensions of the orchard are shown below:

Diagram not to scale



Step 1: Find missing lengths of composite shape:

Missing horizontal = 57 - 25 = 32m; Missing vertical = 62 - 35 = 27m [M1]

Step 2: Split into two rectangles to find total area:

(25 × 62) + (32 × 35) = 1550 + 1120 = 2670 m² [M1 A1]

Step 3: Calculate number of trees (each tree needs 64 m²):

2670 ÷ 64 = 41.718... --> 41 whole trees [M1 A1]

She needs to have an area of at least 64 m² for each fruit tree.

How many fruit trees will she be able to plant in her orchard? (5 marks)

41 trees ✓

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Show your calculations and/or workings out here:

Write your answer in this box:

[End of assessment]

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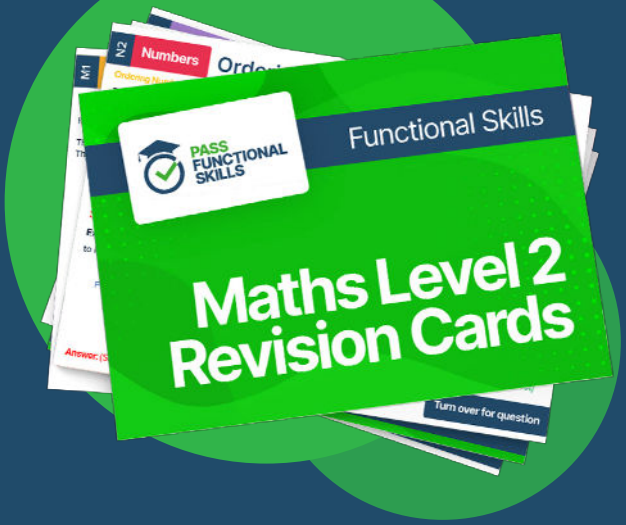

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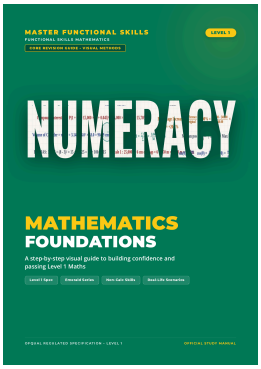

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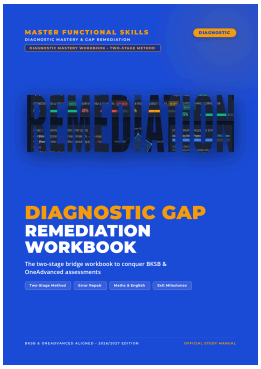

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